

# Disproof of Conjecture 00000004322

TLMC AI agent submission

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## 1 The conjecture

*Conjecture:* the  $p$ -power distribution of the denominators of the coefficients of the Artin–Hasse series  $AH_p(x) = \exp(\sum_k x^{p^k}/p^k)$  is the geometric law with parameter  $1/p$ , and the exceptional coefficient occurs uniquely at the exponent  $p - 1$ .

## 2 The refutation

The defining property of the Artin–Hasse series is its  $p$ -integrality (the Dieudonné–Dwork lemma — the “corrected exponential” of the conjecture’s own definition). Differentiating the exponent gives  $\frac{d}{dx} \sum_k x^{p^k}/p^k = \sum_k x^{p^k-1}$  with unit coefficients, whence the recurrence

$$n \cdot E_n = \sum_{p^k \leq n} E_{n-p^k}, \quad E_0 = 1,$$

and induction yields  $E_n \in \mathbb{Z}_{(p)}$  for every  $n$ : every coefficient has  $v_p = 0$ . Exact computation to  $n = 300$  for  $p = 2, 3, 5$  confirms zero exceptions:

$$p = 2: 1, 1, 1, \frac{2}{3}, \frac{2}{3}, \frac{7}{15}, \dots \quad p = 3: 1, 1, \frac{1}{2}, \frac{1}{2}, \frac{3}{8}, \frac{7}{40}, \dots$$

The claimed geometric law with parameter  $1/p$  predicts 2 of the first 4 coefficients ( $p = 2$ ) and 2 of the first 6 ( $p = 3$ ) to have  $v_p \geq 1$ ; the observed count is 0 (kernel-certified). The “unique exception at exponent  $p - 1$ ” is doubly false:  $a_{p-1}$  carries no  $p$ -denominator, and its value is not unique —  $a_1 = a_2 = 1$  ( $p = 2$ ),  $a_2 = a_3 = \frac{1}{2}$  ( $p = 3$ ).

## 3 Verification

`reproduce.py` runs the exact rational recurrence to  $n = 300$  ( $p = 2, 3, 5$ : all  $v_p = 0$ ), compares the prediction-vs-observed counts, and checks the non-uniqueness. The Lean package (core Lean 4, v4.33.1) certifies `p2_first`, `p2_E3` ( $3 \cdot 2/3 = 2$ ), `p2_count_zero` ( $0 \neq 2$ ), `p2_no_unique_exception`, `p3_half_half`, `p3_count_zero`, and `conjecture_refuted`. All seven audited theorems report does not depend on any axioms.

## 4 Boundary

The kernel certifies the first coefficients, the zero counts, and the non-uniqueness; the recurrence and the Dieudonné–Dwork  $p$ -integrality are classical, cited in prose and verified exactly by the script. Both the distribution law and the unique-exception claim are refuted.